Universitat Pompeu Fabra Barcelona

MTG Music Technology Group

Music Technology Lab S103

Quiz 1: Machine Learning Basics, EnCodec, CLAP, and Dimensionality Reduction

1. Machine learning is about programming explicit rules to detect patterns in data.
False. It is about learning patterns from data rather than relying on explicitly programmed rules.

2. Data, in machine learning, is a collection of examples that a model can use to learn patterns.
True.

3. A model stores learned patterns from data and can always make accurate predictions from new data.
False. After training, a model may fail to generalize well, it can perform well on training data but poorly on unseen data.

4. Training a model adjusts its parameters to better capture patterns in the data.
True.

5. EnCodec compresses audio by learning patterns with a neural network, achieving smaller file sizes than traditional codecs while keeping high quality.
True.

6. CLAP maps audio and text into a common space to match related pairs.
True.

7. CLAP can only recognize sounds it has seen during training and cannot generalize.
False. CLAP is designed to generalize and typically performs well on unseen data.

8. CLAP is trained on large datasets including Freesound to learn patterns between audio and text.
True.

9. Dimensionality reduction helps compress data without loss of information.
False. It usually involves some loss of information in exchange for simpler representations and improved visualization.

10. Dimensionality reduction helps visualize high-dimensional data by preserving patterns in fewer dimensions.
True.